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HT Board Booster

Class X Science

Physics

CBSE Sample Paper Question Bank

Chapter wise Practice

Chapters 1 - 2
Light and Human Eye

Session 2026-27

Practice | Perform | Progress

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Class X Physics
CBSE SQP 2019-2025
Ch- Light

SECTION - A

Q.1. Both a spherical mirror and a thin spherical lens have a focal length of (-)15 cm. What type of mirror and lens are these? **(CBSE SQP 2020-21)**

Q.2. The image formed by a concave mirror is observed to be real, inverted and larger than the object. Where is the object placed? **(CBSE SQP 2020-21)**

OR

Name the part of a lens through which a ray of light passes without suffering any deviation.

Q.3. An object is placed in front of a convex mirror. Its image is formed: **(CBSE SQP 2023-24)**

a) at a distance equal to the object distance in front of the mirror.
b) at twice the distance of the object in front of the mirror.
c) half the distance of the object in front of the mirror.
d) behind the mirror and its position varies according to the object distance.

Q.4. In the following cases, a ray is incident on a concave mirror. In which case is the angle of incidence equal to zero? **(CBSE SQP 2024-25)**

a) A ray parallel to the principal axis.
b) A ray passing through the centre of curvature and incident obliquely.
c) A ray passing through the principal focus and incident obliquely.
d) A ray incident obliquely to the principal axis, at the pole of the mirror.

Q.5. **Assertion (A)**: A convex mirror always forms an image behind it and the image formed is virtual. **(CBSE SQP 2024-25)**

Reason (R): According to the sign convention, the focal length of a convex mirror is positive

Q.6. Arnav was making notes and he wrote down the following statements from his understanding of reflection from curved surfaces. **(CBSE SQP 2025-26)**

I. Concave mirrors can produce both real and virtual images depending on the position of the object.
II. Convex mirrors always produce real, inverted images regardless of the object's position.
III. In both concave and convex mirrors, the image location can be determined using the mirror formula $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ where f is the focal length, v is the image distance, and u is the object distance.

Choose from the following the correct option that lists the correct statements about reflection from curved surfaces.

(a) I and II b) I, II and III c) II and III d) I and III

SECTION - B

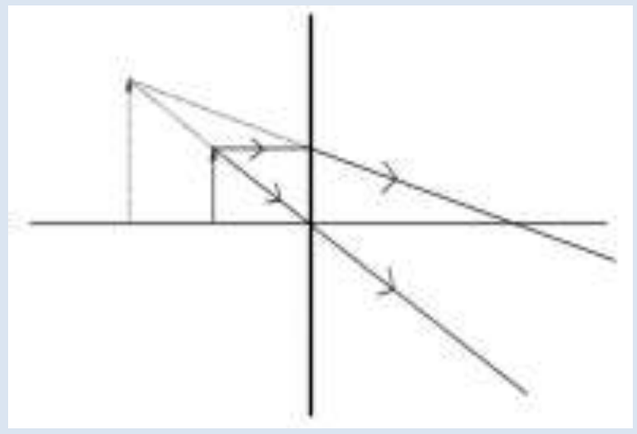
Q.7. The refractive indices of three media are given below: **(CBSE SQP 2023-24)**

Medium	Refractive Index
A	1.6
B	1.8
C	1.5

A ray of light is travelling from A to B and another ray is travelling from B to C.

(a) In which of the two cases the refracted ray bends towards the normal?
(b) In which case does the speed of light increase in the second medium? Give reasons for your answer.

Q.8. The above figure shows the formation of an image by a lens shown by a thick line.



Analyse the figure and answer the following questions.

a) What is the type of lens used?

b) What is the nature of the image?

c) If the image is formed at a distance of 30 cm from the lens and the image is twice the size of the object, then where is the object placed?

(CBSE SQP 2024-25)

SECTION - C

Q.9. Rohit wants to have an erect image of an object using a converging mirror of focal length 40 cm.

(CBSE SQP 2022-23)

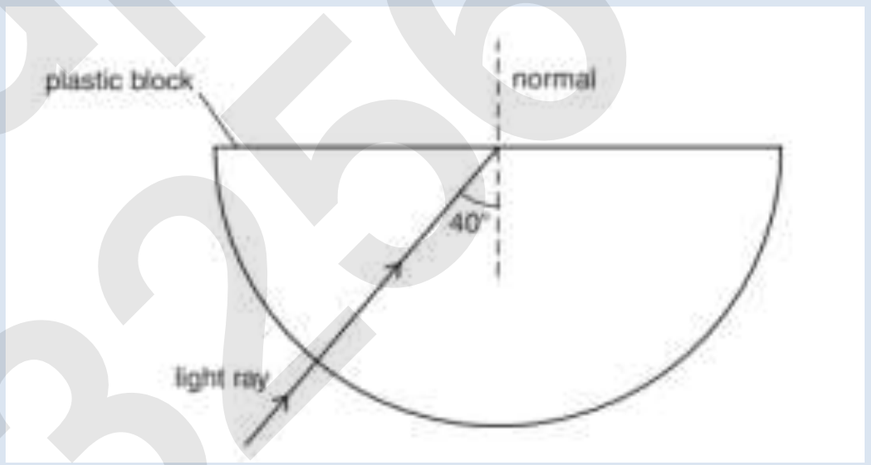
(a) Specify the range of distance where the object can be placed in front of the mirror. Justify.

(b) Draw a ray diagram to show image formation in this case.

(c) State one use of the mirror based on the above kind of image formation.

Q.10. (i) Explain why the refractive index of any material with respect to air is always greater 1.

(ii) In the figure below a light ray travels from air into the semi-circular plastic block. Give a reason why the ray does not deviate at the semi-circular boundary of the plastic block.

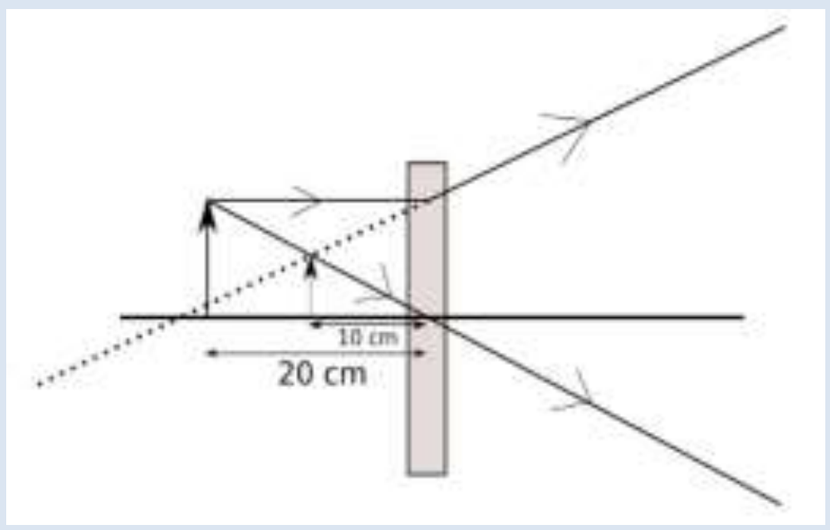


(iii) Complete the ray diagram of the above scenario when the light ray comes out of the plastic block from the top flat end.

(CBSE SQP 2023-24)

Q.11. The above image shows the formation of an image with an optical instrument.

(CBSE SQP 2025-26)



(a) Identify the optical instrument (shown schematically as a rectangle) in the image.

(b) What type of image is formed in this case?

(c) Based on the measurements given in the image, calculate the focal length of the instrument.

Q.12. Refractive index of water with respect to air is 1.33 and that of diamond is 2.42. **(CBSE SQP 2020-21)**

(i) In which medium does the light move faster, water or diamond?

(ii) What is the refractive index of diamond with respect to water?

Q.13. It is desired to obtain an erect image of an object, using concave mirror of focal length of 12 cm. **(CBSE SQP 2019-20)**

(i) What should be the range of the object distance in the above case?

(ii) Will the image be smaller or larger than the object? Draw a ray diagram to show the formation of image in this case.

(iii) Where will the image of this object be, if it is placed 24 cm in front of the mirror?

SECTION - D

Q.14. The above image shows a thin lens of focal length 5m. **(CBSE SQP 2023-24)**



(i) What is the kind of lens shown in the above figure?

(ii) If a real inverted image is to be formed by this lens at a distance of 7m from the pole, then show with calculation where should the object be placed?

(iii) Draw a neatly labelled diagram of the image formation mentioned in (ii)

OR

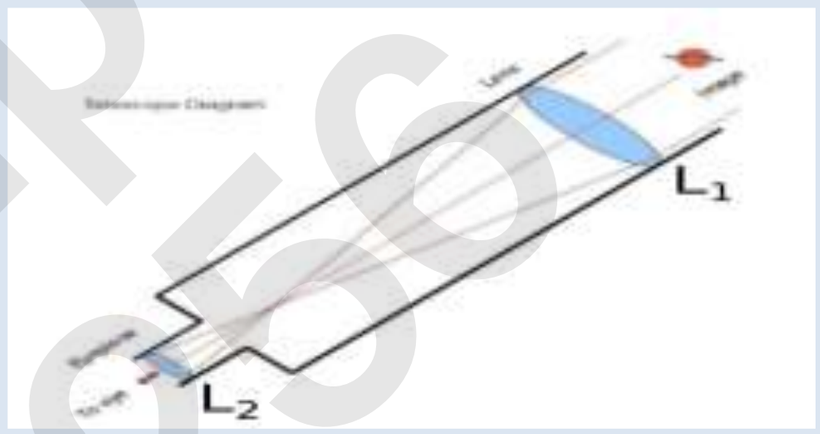
A 10 cm long pencil is placed 5 cm in front of a concave mirror having a radius of curvature of 40 cm.

(i) Determine the position of the image formed by this mirror.

(ii) What is the size of the image?

(iii) Draw a ray diagram to show the formation of the image as mentioned in the part (i).

Q.15. Read the following and answer any four questions from 19 (i) to 19 (v)..



Sumati wanted to see the stars of the night sky. She knows that she needs a telescope to see those distant stars. She finds out that the telescopes, which are made of lenses, are called refracting telescopes and the ones which are made of mirrors are called reflecting telescopes.

So she decided to make a refracting telescope. She bought two lenses, L_1 and L_2. out of which L_1 was bigger and L_2 was smaller. The larger lens gathers and bends the light, while the smaller lens magnifies the image. Big, thick lenses are more powerful. So to see far away, she needed a big powerful lens. Unfortunately, she realized that a big lens is very heavy. Heavy lenses are hard to make and difficult to hold in the right place. Also since the light is passing through the lens, the surface of the lens has to be extremely smooth. Any flaws in the lens will change the image. It would be like looking through a dirty window.

(i) Based on the diagram shown, what kind of lenses would Sumati need to make the telescope? **(CBSE SQP 2020-21)**

a) Concave lenses b) Convex lenses

c) Bi-focal lenses d) Flat lenses

(ii) If the powers of the lenses L_1 and L_2 are in the ratio of 4:1, what would be the ratio of the focal length of L_1 and L_2?

a) 4:1 b) 1:4 c) 2:1 d) 1:1

(iii) What is the formula for magnification obtained with a lens?

a) Ratio of height of image to height of object b) Double the focal length.

c) Inverse of the radius of curvature. d) Inverse of the object distance

(iv) Sumati did some preliminary experiment with the lenses and found out that the magnification of the eyepiece (L_2) is 3. If in her experiment with L_2 she found an image at 24 cm from the lens, at what distance did she put the object?

a) 72 cm b) 12 cm c) 8 cm d) 6 cm

(v) Sumati bought not-so-thick lenses for the telescope and polished them. What advantages, if any, would she have with her choice of lenses?

a) She will not have any advantage as even thicker lenses would give clearer images.

b) Thicker lenses would have made the telescope easier to handle.

c) Not-so-thick lenses would not make the telescope very heavy and also allow considerable amount of light to pass.

d) Not-so-thick lenses will give her more magnification

SECTION - E

Q.16. The above images are that of a specialized slide projector. Slides are small transparencies mounted in sturdy frames ideally suited to magnification and projection, since they have a very high resolution and a high image quality. There is a tray where the slides are to be put into a particular orientation so that the viewers can see the enlarged erect images of the transparent slides. This means that the slides will have to be inserted upside down in the projector tray.



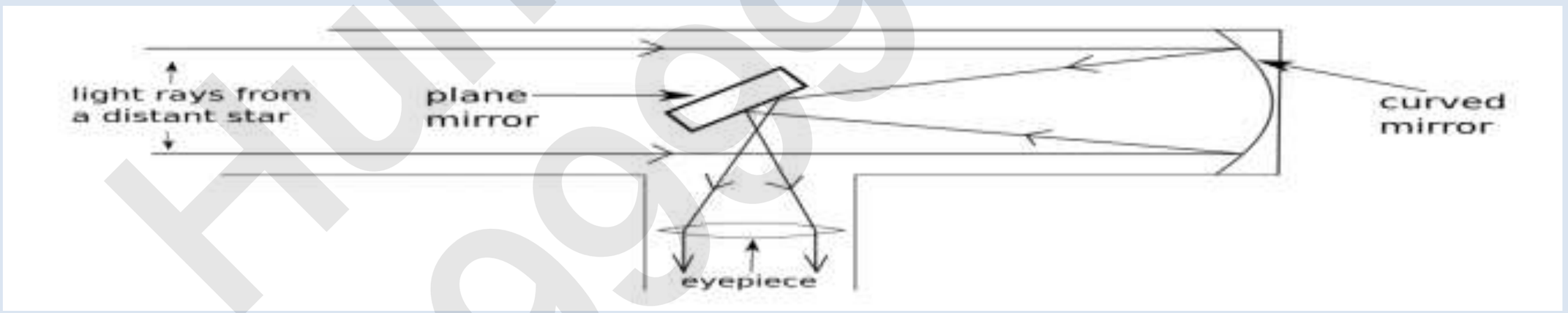
To show her students the images of insects that she investigated in the lab, Mrs. Iyer brought a slide projector. Her slide projector produced a 500 times enlarged and inverted image of a slide on a screen 10 m away. **(CBSE SQP 2022-23)**

(a) Based on the text and data given in the above paragraph, what kind of lens must the slide projector have?

(b) If v is the symbol used for image distance and u for object distance then with one reason state what will be the sign for in the given case?

(c) A slide projector has a convex lens with a focal length of 20 cm. The slide is placed upside down 21 cm from the lens. How far away should the screen be placed from the slide projector's lens so that the slide is in focus? OR

(d) When a slide is placed 15 cm behind the lens in the projector, an image is formed 3 m in front of the lens. If the focal length of the lens is 14 cm, draw a ray diagram to show image formation. (not to scale).



Q.17. The above image is that of a reflecting telescope. Reflecting telescopes revolutionized our ways of looking into the sky. They employ mirrors to gather and focus light, rather than relying solely on lenses as in their refracting counterparts. These telescopes utilise precisely shaped and polished mirrors to capture incoming light and reflect it to a focal point, where it forms an image for observation. **(CBSE SQP 2024-25)**

a) What kind of image of the star is seen by the observer at the eye piece?

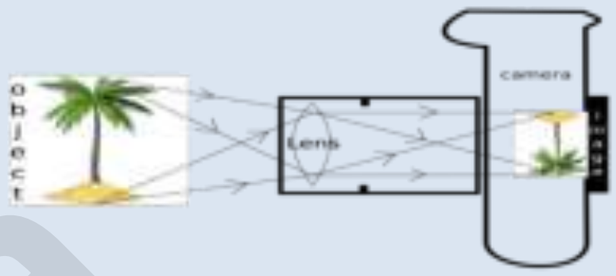
b) What kind of mirror is used in this reflecting telescope?

c) Explain with reason what kind of optical device (type of lens or mirror) that is used at the eyepiece. **OR**

d) What is the role of the plane mirror in the telescope?

Q.18. The above image is that of a Digital Single Lense Reflector (DSLR) Camera which are used to take high resolution photographs by professional photographers. The second image of the above two is a schematic diagram of how an image is formed on the sensor of the camera. Based on your understanding of the lenses, answer the following questions.





a) What type of lens is used in the DSLR camera shown in the image?

b) What type of image is formed on the sensor?

Attempt either subpart C or D.

c) A photographer is using a DSLR camera with a lens of focal length f=50 mm to take a close-up photograph of a small object. The lens projects an image onto the camera sensor that is located 60 mm behind the lens. Calculate the object distance (i.e., the distance between the object and the lens).

OR

d) A photographer is using a DSLR camera to take a picture of a flower. The flower is positioned 150 mm away from the camera lens. The actual height of the flower is 80 mm, and the image height formed on the camera's sensor is measured to be 20 mm. Calculate the focal length of the camera lens. **(CBSE SQP 2025-26)**

Q.19.

(i) A lens produces a magnification of -0.5. Is this a converging or diverging lens? If the focal length of the lens is 6 cm, draw a ray diagram showing the image formation in this case. **(CBSE SQP 2019-20)**

(ii) A girl was playing with a thin beam of light from a laser torch by directing it from different directions on a convex lens held vertically. She was surprised to see that in a particular direction, the beam of light continues to move along the same direction after passing through the lens. State the reason for her observation. Draw a ray diagram to support your answer.

OR

(i) On entering in a medium from air, the speed of light becomes half of its value in air. Find the refractive index of that medium with respect to air?

(ii) A glass slab made of a material of refractive index n1 is kept in a medium of refractive index n_2.
A light ray is incident on the slab. Draw the path of the rays of light emerging from the glass slab, if (i) $n_1 > n_2$ (ii) $n_1 = n_2$ (iii) $n_1 < n_2$

Class X Physics
CBSE SQP 2019-2025
Ch- Human Eye

Section - A

Q.1. When we enter a dark room coming from outside, immediately the things inside the room do not appear clear to our eyes. This is because **(CBSE SQP 2019-20)**

(i) pupils do not open at all in the dark.
(ii) pupils take time to adjust.
(ii) light travels slower in a dark room.
(iv) pupils open very quickly in the dark.

Q.2. The phenomena of light responsible for the working of the human eye is **(CBSE SQP 2019-20)**

(i) reflection
(ii) refraction
(iii) power of accommodation
(iv) persistence of vision

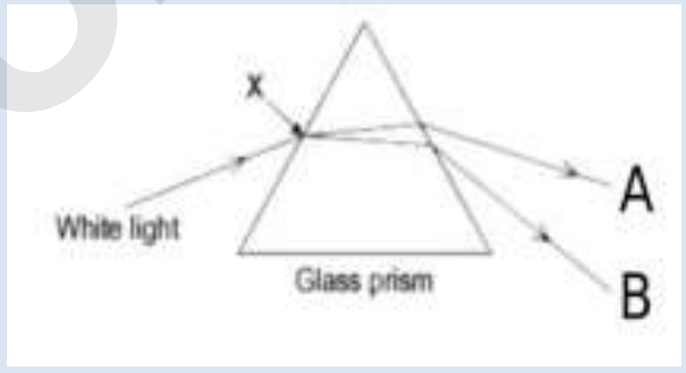
Q.3. Why does the Sun appear white at noon? **(CBSE SQP 2020-21)**

Q.4. When light enters the atmosphere it strikes on extremely fine particles, which deflect the rays of light in all possible directions, This is due to - **(CBSE SQP 2023-24)**

(a) reflection of light
(b) atmospheric refraction
(c) scattering of light
(d) dispersion of light

Q.5. Choose the correct option for the colour of rays for A and B. **(CBSE SQP 2024-25)**

	Colour of Ray A	Colour of Ray B
A	Blue	Red
B	Green	Yellow
C	Red	Violet
D	Violet	Indigo

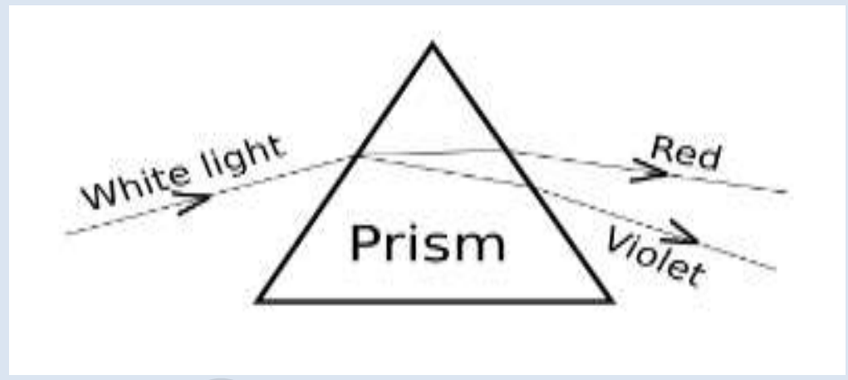


Q.6. Choose the correct option from the below which explains the reason for us to perceive the day sky as blue. **(CBSE SQP 2024-25)**

(a) As sunlight passes through the atmosphere, shorter wavelengths, such as blue are scattered more than other colors.
(b) The sky appears blue because all colors are scattered equally, but blue light is stronger and more visible to the human eye.
(c) The blue color of the sky is due to longer wavelengths like red and orange scattering more than shorter wavelengths, making blue stand out more.
(d) The atmosphere contains blue-colored particles that give the sky its blue appearance.

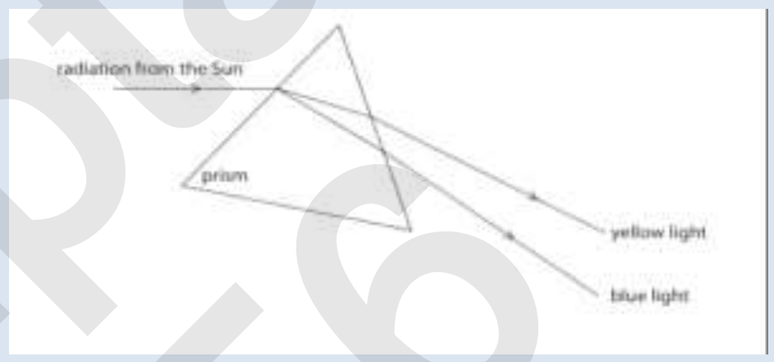
Section - B

Q.7. A student observes the above phenomenon in the lab as a white light passes through a prism. Among many other colours, he observed the position of the two colours Red and Violet.



What is the phenomenon called? What is the reason for the violet light to bend more than the red light? **(CBSE SQP 2020-21)**

Q.8. State the phenomena observed in the above diagram. Explain with reference to the diagram, which of the two lights mentioned above will have the higher wavelength?

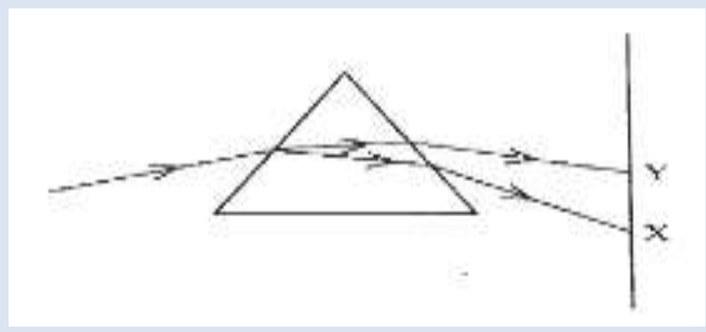


OR

How will you use two identical prisms so that a narrow beam of white light incident on one prism emerges out of the second prism as white light? Draw the diagram. **(CBSE SQP 2022-23)**

Section –C

Q.9. In the figure given below, a narrow beam of white light is shown to pass through a triangular glass prism. After passing through the prism, it produces a spectrum XY on the screen. **(CBSE SQP 2019-20)**



(i) Name the phenomenon.
(ii) State the colours seen at X and Y.
(iii) Why do different colours of white light bend at different angles through a prism?

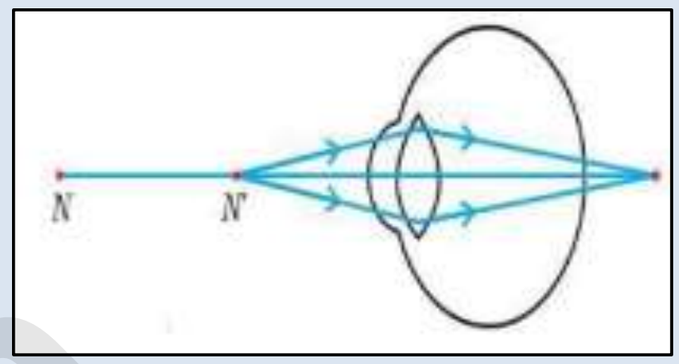
Q.10. (i) What is visible spectrum?
(ii) Why is red used as the stopping light at traffic signals?
(iii) Two triangular glass prisms are kept together connected through their rectangular side. A light beam is passed through one side of the combination. Will there be any dispersion? Justify your answer. **(CBSE SQP 2019-20)**

Q.11. A lens of focal length 5 cm is being used by Debashree in the laboratory as a magnifying glass. Her least distance of distinct vision is 25 cm. **(CBSE SQP 2022-23)**

(i) What is the magnification obtained by using the glass?
(ii) She keeps a book at a distance 10 cm from her eyes and tries to read. She is unable to read. What is the reason for this?
(b) Ravi kept a book at a distance of 10 cm from the eyes of his friend Hari. Hari is not able to read anything written in the book. Give reasons for this?

Q.12. (a) What is the fundamental difference between hypermetropia and myopia 3 in terms of the optical experience of a person? **(CBSE SQP 2024-25)**

(b) The diagram below shows a special case of an eye defect.

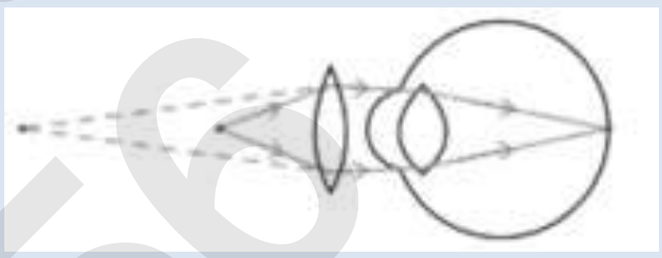


(i) What is the defect that is shown in the figure?

(ii) State one cause for such a defect?

(iii) Explain with reason if a concave lens can be used to correct the defect.

Q.13. The above image shows a corrective measure for a particular defect of vision.
(CBSE SQP 2025-26)



(i) Identify the defect of vision and state what kind of lens is used to correct this deficiency.

(ii) Draw and label a ray diagram that shows the defect of vision in the above case before correction.

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